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Using a CNN-LSTM for basic behaviors detection of a single dairy cow in a complex environment

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ABSTRACT

The basic behaviors of dairy cows (drinking, ruminating, walking, standing and lying) are closely related to their physiological health status. Consequently, intelligent behavior recognition is of significance for the automatic diagnosis and precision farming of dairy cows. Realizing the accurate behaviors classification in complex environments involving low quality surveillance videos, complex illumination and weather changes is a key problem in dairy farming that must be solved. In this study, CNN-LSTM (fusion of convolutional neural network and long short-term memory) an algorithm for recognizing the basic behaviors of a single cow, was proposed. First, the VGG16 trained on ImageNet was used as the network skeleton to extract the feature vector sequence corresponding to each video, so as to avoid the shortcomings of traditional feature engineering which were time-consuming and laborious. Then, these features were input Bi-LSTM (bidirectional long short-term memory) classification model, which could extract semantic information of time series data in two directions, so as to realize accurate recognition of dairy cow's basic behaviors. To verify the effectiveness of the VGG16 feature extraction network used in this research, 1370 segments of approximately 18 h of videos collected from dairy farm monitoring cameras were tested and compared with those of five different feature extraction networks based on VGG19, ResNet18, ResNet101, MobileNet V2 and DenseNet201. Moreover, the effects of changes in illumination, weather, and wind velocity on behaviors recognition were tested and discussed. The test results indicated that the *precision* of the proposed algorithm for the recognition of the five behaviors ranged from 0.958 to 0.995, the *recall* ranged from 0.950 to 0.985, and the *specificity* ranged from 0.974 to 0.991, while the average *precision, recall* and *specificity* were 0.971, 0.965 and 0.983, respectively. The average recognition *accuracy* of the proposed method was 0.976, which was higher than the methods based on VGG19, ResNet18, ResNet101, MobileNet V2 and DenseNe201 by 0.08×10^{-2}, 1.97×10^{-2}, 2.19×10^{-2}, 2.85×10^{-2} and 2.34×10^{-2}, respectively. Furthermore, the influences of illumination, weather and wind speed on the algorithm were discussed. The results showed that the difference of behavior recognition accuracy of this method under the above interference was less than 0.02, indicating that the method is good in stability. The research results showed that it was feasible to use the proposed algorithm to recognize behaviors of a single target dairy cow. This study could not only provide valuable references for the behaviors identification and understanding of multiple target dairy cows based on computer vision in complex environments such as low-quality surveillance video, complex illumination and weather variations, but also contribute to their physiological health assessment and remote diagnosis. The study may be valuable for the dairy cows' prevention and treatment of health and reproduction problems using the "medical-engineering interdisciplinary" approach.

1. Introduction

The behavior of a dairy cow is the comprehensive embodiment of its physiological activity states. Identifying basic behaviors (drinking, ruminating, walking, standing and lying) is not only helpful for the physiological health evaluation and disease treatment of dairy cows but

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also important to realize precision farming (Sepúlveda-Varas et al., 2014; Gu et al., 2017; Yin et al., 2020). The existing behaviors recognition method of dairy cows, which relied on human eye observation and experience judgment, was difficult to meet the increasing demand of the cultivation scale due to the problems of high labor cost, work intensity, and fatigue rate. The automatic identification of basic behaviors is key to diagnose diseases, improve the farm's economic benefits and reduce the culling rate of dairy cows (Paudyal et al., 2018; Song et al., 2018; Grimm et al., 2019).

Considerable researches on the recognition of cow behaviors have been carried out and achieved innovative results. These studies could be divided into two types: contact-sensor and non-contact-sensor-based methods (Hoang et al., 2018; Wang et al., 2018; Barker et al., 2018; Pratama et al., 2019). Techniques based on contact sensors mainly collect different behavior movements through sensors fixed on cows and recognize behaviors according to the difference of different behavior's movement or acoustic data (Abell et al., 2017; Kiani, 2018; Shen et al., 2020; Tamura et al., 2019; Benaissa et al., 2019; Aydin and Berckmans, 2016; Chelotti et al., 2020). Although the contact sensors have a high precision, they are easy to cause cow stress and would be affected by factors such as scraping and moisture infiltration, which may damage components and affect the service life of detection devices. In contrast, the computer vision technology has a low cost and high response speed, which could avoid stress problems caused by contact-sensor-based methods. Consequently, it is an important development direction in the dairy cow behaviors recognition (He et al., 2016a).

Computer vision, as a non-contact, stress-free, and cost-effective technology, is the main trend for animal behaviors recognition (Nasirahmadi et al., 2017; Stewart et al., 2017; Zhu et al., 2017; Aydin, 2017; Jiang et al., 2019; Wu et al., 2020; Zhu et al., 2020). Lee et al. (2016) proposed a method for identifying the attack behaviors among pigs. Kinect devices were used to obtain the activity characteristics of the pigs in a pigsty, and SVM (support vector machine) were used to identify the attack behaviors. By identifying 330 segments of pig videos and comparing the results with those of manual observation, it was observed that the accuracy of the method to identify the pig attack behavior could reach 0.90. Chen et al. (2020) proposed a pig's drinking and drinker-playing behavior recognition method based on ResNet50 convolutional neural network and LSTM, and the classification accuracy in the body and head regions of interest was 0.87 and 0.93, respectively. Måløy et al. (2019) proposed a dual-stream recurrent network (DSRN) based on a convolutional neural network to identify Feeding and NonFeeding behavior of salmon in a fish farm. The test results showed that the prediction accuracy of this method could reach 0.80. Porto et al. (2015) modeled and verified the feeding and standing behavior detector of dairy cows by designing a method based on Viola-Jones algorithm and using a multi-camera video recording system. The sensitivity of this system to feeding and standing behaviors was about 0.87 and 0.86, respectively, which indicated that the method could recognize standing and eating behaviors well. Guo et al. (2019) proposed a cow mounting behavior recognition method based on the region geometry and optical flow characteristics. By removing the irrelevant background through the use of the mask operation, the inter-frame difference and background subtraction were used to extract the cow target region characteristics. Finally, the support vector machine was used to recognize the cow climbing behavior. The test results pertaining to 30 videos showed that the recognition accuracy of this method was 0.98.

In the above mentioned noncontact livestock behavior recognition methods, the use of depth cameras could achieve a satisfactory recognition effect; however, most of the monitoring videos in dairy farms are still captured using ordinary RGB cameras. Although some achievements have been made in the identification of the lying and estrus behaviors of dairy cows, these approaches are mainly aimed at identifying certain behaviors, and most of them require the manual design of the feature engineering. This process is relatively complex and vulnerable to environmental changes (Zhao and Liu, 2019). In the multibehavior recognition research, the four basic behaviors of calves were mainly studied during the day, and the video materials were high resolution videos recorded by the researchers using cameras; the behavior recognition problems in complex environments such as low-quality video, complex illumination and weather variations were not considered.

With the rapid development of computer hardware and computing power, deep learning and computer vision are increasingly used in intelligent monitoring research of dairy farming (Guzhva et al., 2018; Cockburn, 2020). To solve the above-mentioned problems, the CNN-LSTM algorithm was proposed to recognize basic behaviors of cows in a complex environment in this study. Firstly, the VGG16 convolution neural network was used to extract the feature vector sequence of the videos collected using the low-quality monitoring camera in the cattle farm, so as to overcome the shortcomings of the traditional feature engineering, such as the time and labor intensiveness. Furthermore, the designed Bi-LSTM classification network model was used to accurately recognize basic behaviors (drinking, ruminating, walking, standing, lying) of dairy cows in a complex environment. This study could not only provide valuable technical reference for dairy cow's basic behavior recognition in a complex environment, but also lay a technical foundation for remote physiological health assessment and disease diagnosis.

2. Materials and methods

2.1. Materials

The test videos used in this research were captured on the outdoor exercise area of Keyuan Dairy Farm in Shaanxi Yangling, China from 6: 00 a.m. to 12: 00 p.m. every day from September 10 to November 24, 2019. The subjects were 31 Holstein cows (body length 2.3–2.6 m, body weight 650–700 Kg) in the middle lactation period who entered the outdoor exercise area after eating from the cowshed. To enhance the practicability of the proposed method, all the videos were collected from the remote video acquisition platform in dairy farms that we have built, rather than high-definition videos shot manually. As shown in Fig. 1, it is a framework of the remote video acquisition platform. Among them, the infrared surveillance camera was used to collect video and the wireless bridge was used for video transmission. The internet server accesses the Internet through ADSL (Asymmetric Digital Subscriber Line) and configures DDNS (Dynamic Domain Name Server). The remote monitoring platform obtained the server IP (Internet Protocol) through analysis and accesses the server. Then the server assigned the virtual subnet IP to the monitoring platform. Thus, the camera arranged in the dairy farm could be monitored on the remote monitoring platform, and the camera could be controlled to rotate, and parameters such as focal length, aperture, etc. could also be set. In this study, the wind velocity in each test video was obtained from the meteorological observation platform of the dairy farm according to the shooting date.

The schematic of the video acquisition process is shown in Fig. 2. Dairy cows would exercise or drink water in the outdoor exercise area after eating in the cowshed, where the drinking pool is a rectangular pool (3.0×1.0 m). A Hikvision DS-2DM1-714 dome webcam (Hikvision Digital Technology Co. Ltd., Hangzhou, China) was installed at the outer corner of the outdoor exercise area to capture test videos, and the camera was approximately 3 m vertically above the ground. Fig. 3 is the monitoring platform. The camera's focal length was adjusted firstly on the operating panel to ensure that only single cow was included in the screen, and then the record button was clicked to record test videos, which made it unnecessary to identify the cow target when studying behaviors recognition.

A total of 4566 videos (approximately 63 h) of 5 basic behaviors (drinking, ruminating, walking, standing and lying) were captured from the outdoor exercise area. The duration of each video was 10–55 s, and the frame rate was 25 fps. In addition, each video had only one behavior label. The reason why feeding behavior was not considered was that cows usually exercise and drink water in the outdoor exercise area after

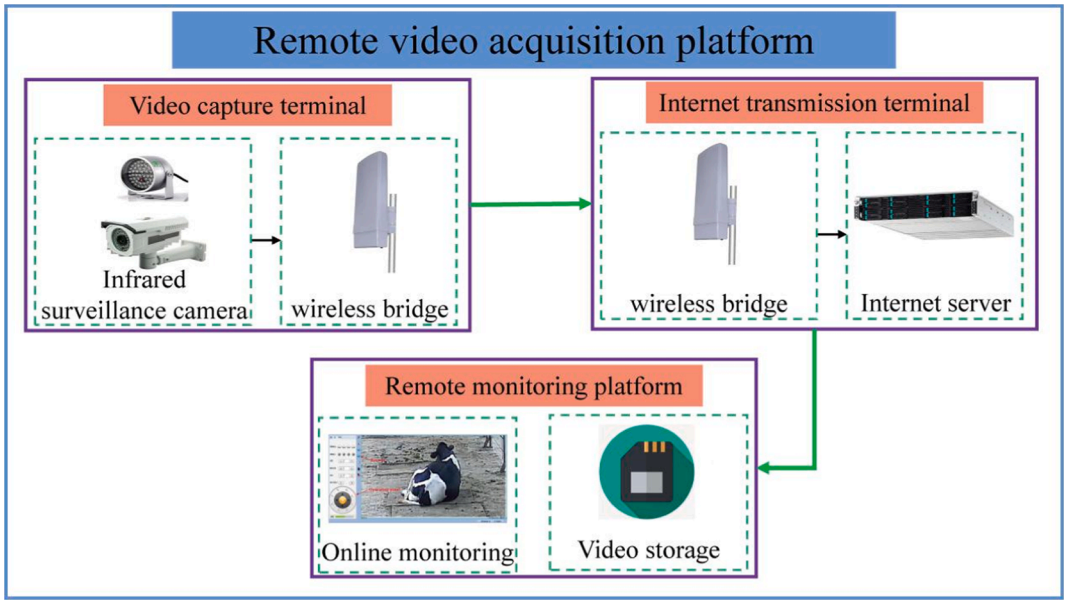


Fig. 1. Framework of remote video capture platform.

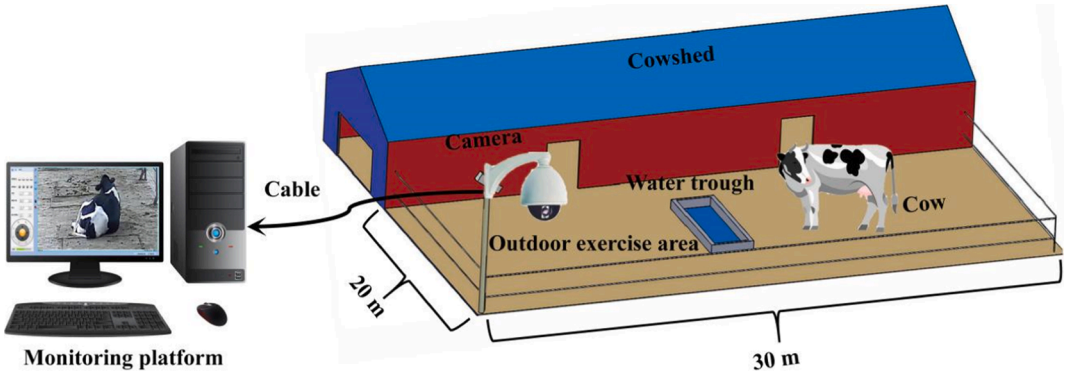


Fig. 2. Schematic diagram of video capture process.



Fig. 3. Interface of monitoring platform.

feeding in the cowshed, making the monitoring camera installed in the outdoor exercise area unable to capture feeding behavior. These 4566 videos were compiled into a dataset according to the format of the commonly used hdbm51 dataset in behavior recognition. Subsequently, 70.00% (approximately 45 h) of the dataset was used as the training data for training the cow behavior recognition model and 30.00% (approximately 18 h) of the dataset was used as the test data for testing the model (Gu et al., 2017; He et al., 2016b). The imbalance between classes meat that the proportion of each class in the data set was unbalanced, and the number of some classes may be several times or even dozens of times that of other classes. This difference was normal and acceptable within an order of magnitude, so data imbalance was not considered in this study (Ye and Ding, 2018; Ren et al., 2017). The definitions of the basic behaviors data set are shown in Table 1 below.

The dataset in this study has the following characteristics: For the target selection, the videos used in this study were collected from 31 different cow targets, which ensured the adaptability of the method to different cow targets. For the illumination change, the video acquisition time was from 6: 00 a.m. to 12: 00 p.m. In this manner, the dataset contained various illumination changes, which ensured the adaptability of this method to different illumination conditions. In addition, the test videos included those collected in sunny and rainy days, which could help verify the robustness of the method to weather changes. Fig. 4 shows a schematic of the different behaviors included in the dataset.

The processor of the video processing platform was an Intel Xeon E3-1620, with a 3.50 GHz processor, 32 GB RAM, 500 GB hard disk, and 11 GB NVIDIA RTX 2080Ti GPU. The algorithm was developed using Microsoft Visual Studio 2015 and MATLAB 2019b.

2.2. Basic behaviors recognition using CNN-LSTM

2.2.1. Overall technical approach

To recognize the basic behaviors in a complex environment (low-quality surveillance video, complex lighting, variable weather, etc.), the technical route adopted in this study was shown in Fig. 5. The proposed algorithm mainly included two parts. The VGG16 framework was firstly used as the network skeleton to extract the feature sequence corresponding to each video, to effectively avoid the complexity and sensitivity of the traditional feature engineering design process to environmental changes. The second part was the basic behaviors recognition. The feature vector sequence corresponding to the test video extracted by VGG16 was sent to the designed Bi-LSTM model, which integrated the forward and backward LSTM to capture the hidden information of the past and the future, and combined the two parts to form the final output to realize the recognition of dairy cow's basic behaviors.

2.2.2. Video feature vector sequence extraction based on VGG16

The test videos were collected from monitoring cameras with a low resolution and included low illumination videos recorded at night. Convolution neural network mainly relied on convolution layer to extract rich features through convolution operation. Furthermore, pooling operation made the extracted features have better translation invariance, which made the features extracted by CNN richer and more scientific than simple manual features. To avoid the influence of the complex environmental changes on the traditional feature engineering and improve the robustness of the behaviors recognition algorithm, the video feature vectors were extracted using the convolution neural network (Hinton and Salakhutdinov, 2006; Mabrouk and Zagrouba, 2018).

VGG network was a deep convolution neural network structure proposed by Simonyan and Zisserman of the Oxford University. By successively using multiple 3×3 convolutions to characterize complex features, this network could not only simulate the effect of larger receptive fields, increase the number of feature map channels, but also achieve better extraction (Simonyan and Zisserman, 2014). As shown in Fig. 6, the VGG16 network trained in ImageNet data set was used to extract the feature vector sequence of the test video in this study. After the input video was decomposed into sequence frames, the features were extracted and optimized through 13 convolution layers and 5 pooling layers in VGG16. The feature extraction was performed by removing the Fc8 layer and all the subsequent layers in the VGG16 network, and then the feature map of each frame was transformed into a line of feature vectors, so the test video would be transformed into a feature vector sequence after feature extraction. The reason why fc7 layer of the VGG16 network was used to extract features in this study was that the dimension of the convolution features was high, and it would consume a lot of memory and time to extract video features (Yuan et al., 2019; Huang et al., 2016).

The reason why we chose to extract features in the fully connected layer was that the leakage regularization was adopted in VGG16, which made the extracted features meet Fisher criterion and were more conducive to classification and recognition tasks (Jin and Dong, 2016).

2.2.3. Recognizing the basic behaviors of dairy cows based on Bi-LSTM

The behaviors of dairy cows were closely related to time, and it was difficult to process time series data with a general neural network. The Long Short-Term Memory (LSTM) is a special recurrent neural network (RNN). Each LSTM unit has three gates to control the transmission of information, and it could learn the long term dependence to solve the problems of gradient disappearance and explosion (Hochreiter and Schmidhuber, 1997). The characteristic of the LSTM is that the current cell obtains the information of all the cells before a given cell. However, it cannot obtain the information of the cells after this cell. The Bi-LSTM is an improved version of the common LSTM. The basic idea is to use a forward and backward LSTM to capture the information implied in the past and future, respectively. Consequently, this model could better process the time series data.

The structure of Bi-LSTM is shown in Fig. 7. The first LSTM layer was used to calculate the sequence information of the current time point, while the second was used to reverse read the same sequence and add reverse sequence information, so as to enhance its video semantic feature extraction ability. Hidden outputs between LSTM layers were passed not only to neighboring units, but also to the input of the next LSTM layer. Since there was no interaction between neurons in the two states, the network weights could be updated by forward and backward propagation of neurons.

After extracting the feature vector sequence of each video, a Bi-LSTM classification model was constructed to recognize basic behaviors. As shown in Fig. 8, the constructed Bi-LSTM model was utilized firstly to extract the features from the feature vector sequences, and the learning result was later employed as the input of the fully connected layer. Finally, the Softmax layer was used to classify and output the behaviors recognition result. To avoid overfitting and improve the generalization ability of the model, the dropout layer was added after the Bi-LSTM layer.

The main steps to recognize cow behaviors based on the Bi-LSTM were as follows:

Table 1
Definition of basic behaviors data set.

Behavior	Description	Number of videos
Drinking	Stand by the water tank with its mouth in the tank	785
Ruminating	Mouth chews repeatedly without other vigorous movement	911
Walking	Legs move repeatedly and cow position changes greatly	878
Standing	Legs remain vertical without other strenuous movement	940
Lying	Abdomen clings to the ground without other strenuous movement	1052

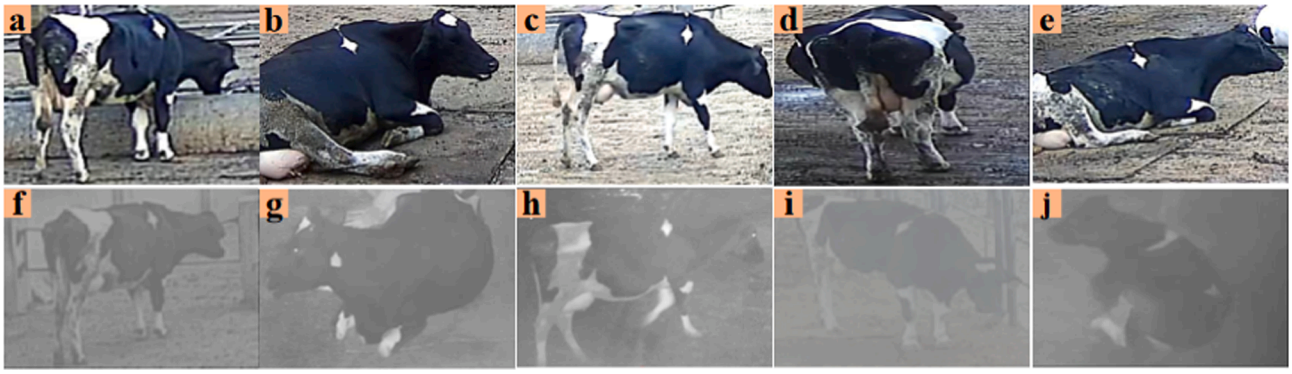


Fig. 4. Different behaviors of dairy cows: (a), (b), (c), (d) and (e) are respectively drinking, ruminating, walking, standing lying behavior during the day, while (f), (g), (h), (i) and (j) are these kinds of behaviors captured at night, respectively.

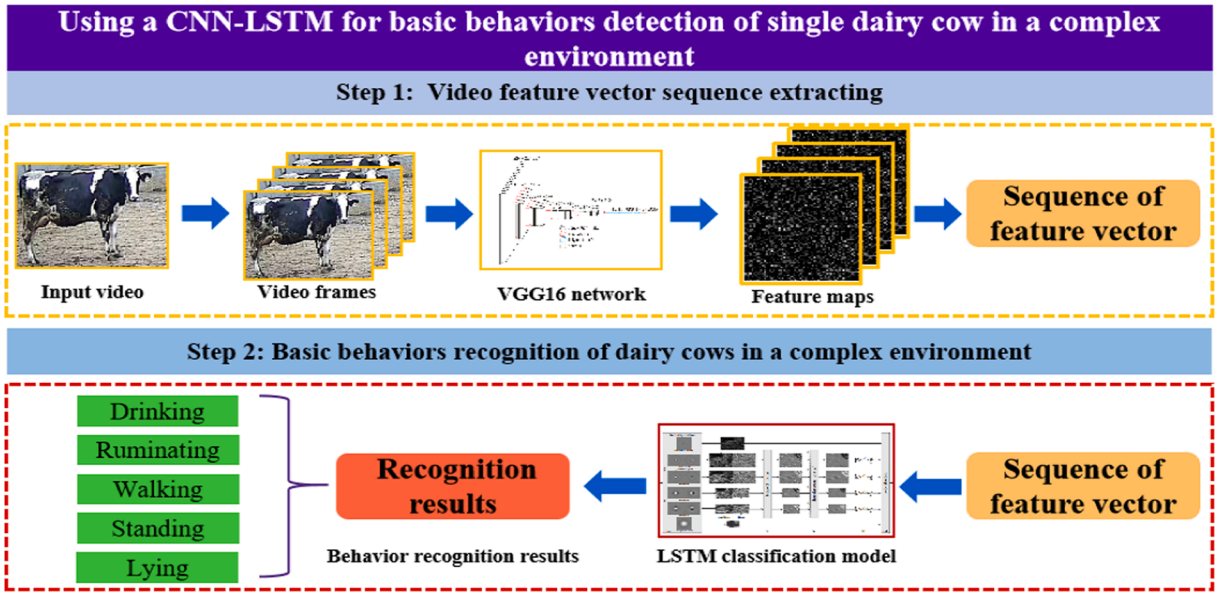


Fig. 5. Technical route of basic behaviors recognition of dairy cow.

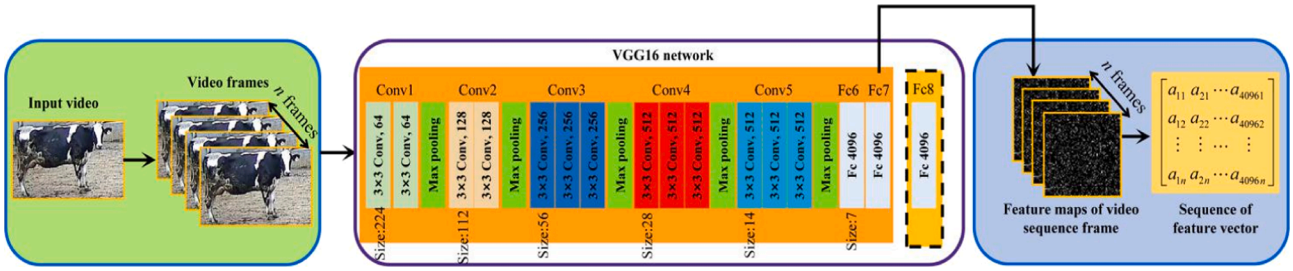


Fig. 6. Process of video feature vector sequence extraction based on VGG16.

(1) Data organization.

By extracting the feature vector sequence of each video, the data set was first divided into a training set (3196 videos) and test set (1370 videos) in a ratio of 7:3. Next, the feature vector sequence corresponding to each video was extracted, and the labels were established for different behaviors. Finally, the feature vector sequence of each video was normalized.

(2) Design and training of the Bi-LSTM classification model.

The model parameters were set, and the training set was used to train the Bi-LSTM model. The main model parameters included the input size, number of hidden units, batch size and dropout rate for each round of training.

Since the feature vector sequence of the video constructed in this study was a row vector extracted and converted from the Fc7 layer, and the output size of the Fc7 layer was 4096, the input size of the Bi-LSTM model was set as 4096. The model parameters are listed in Table 2. In the process of designing the Bi-LSTM model, the number of hidden units considerably influenced the classification performance of the modules. When 256 hidden units were used, the *Dropout_rate* was 0.5 and the

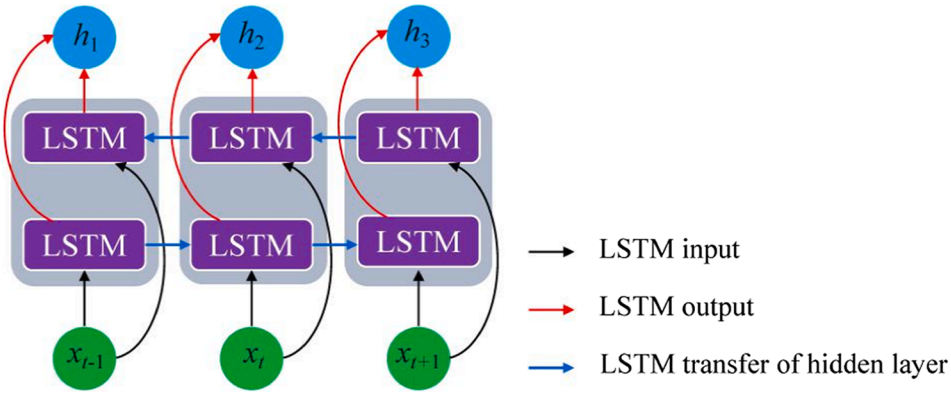


Fig. 7. Schematic diagram of Bi-LSTM structure.

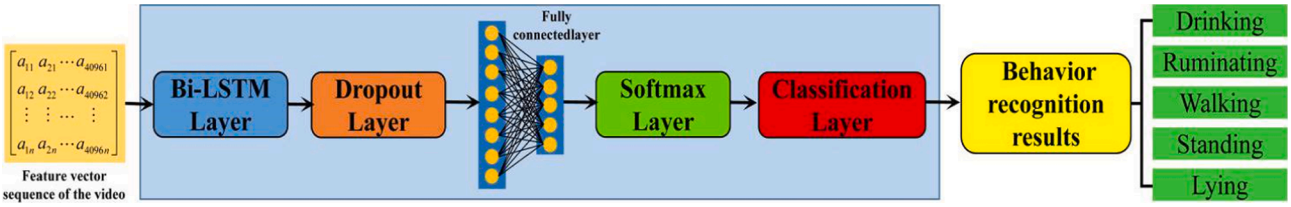


Fig. 8. Behaviors recognition model based on Bi-LSTM.

Table 2
Parameters setting of Bi-LSTM classification model.

Parameter name	Value
Input size	4096
Initial learning rate	1.00×10^{-4}
Batch_size	20
Dropout rate	0.5
Iteration	7155
Gradient descent strategy	Adam

Table 3
. Parameters setting of Bi-LSTM classification model.

Authors	Behavior classes	Sensor type	Recognition result
Kiani	Standing, lying down, Walking and looking for the grass	A accelerometer fixed on leg	0.94
Smith et al.	Grazing, walking, ruminating, resting, and "Other"	A inertial measuring unit (IMU) fixed on neck	0.46–0.98
Wang et al.	Feeding, lying, standing, lying down, standing up, normal walking, and active walking	Accelerometer and position sensor fixed on the leg	≧0.80
Chelotti et al.	Ruminating, grazing, and "Other"	A microphone fixed on the forehead	≧0.76
Tsai et al.	Drinking	A Raspberry Pi Camera V2 module	≧0.90
Guo et al.	Mounting	A handheld HD camera fixed outside the fence	0.98
Simona et al.	Feeding, and standing	Multi-camera fixed in cowshed	≧0.86
He et al.	Lying, standing, walking, and running	A handheld HD camera fixed outside the fence	0.95
Ours	Drinking, ruminating, walking, standing, and lying	A low-quality surveillance camera	0.98

Batch_size was 20, the training accuracy for the training set was the highest. Among them, the regularization effect was the most obvious when the *Dropout_rate* was set to 0.5, and the *Batch_size* was mainly depended on the processor (Srivastava et al., 2014; Baldi and Sadowski, 2013) (see Table 3).

The training set and test set were used to train and test the model, respectively. As shown in Fig. 9a and b, which respectively represent the precision and loss curves in the training process. It could be seen that the convergence speed of the training curve was higher in the initial stage of training the model, indicating that the learning efficiency of the model was higher. With the deepening of the model training, the slope of the training curve gradually decreased. Finally, when the training iteration reached approximately 7000, the model learning efficiency gradually reached saturation and presented slight fluctuations.

2.3. Evaluation criterion

To verify the recognition accuracy of the proposed algorithm for different behaviors, four evaluation indices of *accuracy*, *precision*, *recall* and specificity were used to evaluate the model performance. The *accuracy*, *precision*, *recall* and specificity were calculated as shown in Eqs. (1)–(4):

$$accuracy = \frac{TP + TN}{FP + FN + TP + TN} \tag{1}$$

$$precision = \frac{TP}{TP + FP} \tag{2}$$

$$recall = \frac{TP}{TP + FN} \tag{3}$$

$$specificity = \frac{TN}{FP + FN} \tag{4}$$

Here, the *accuracy* is the ratio of the number of correctly recognized test videos to the total number of test videos, *precision* is the ratio of the number of videos for a behavior in the test set to the number of videos for that behavior in the recognition results, *recall* is the ratio of the number of videos correctly identified for a behavior to the total number

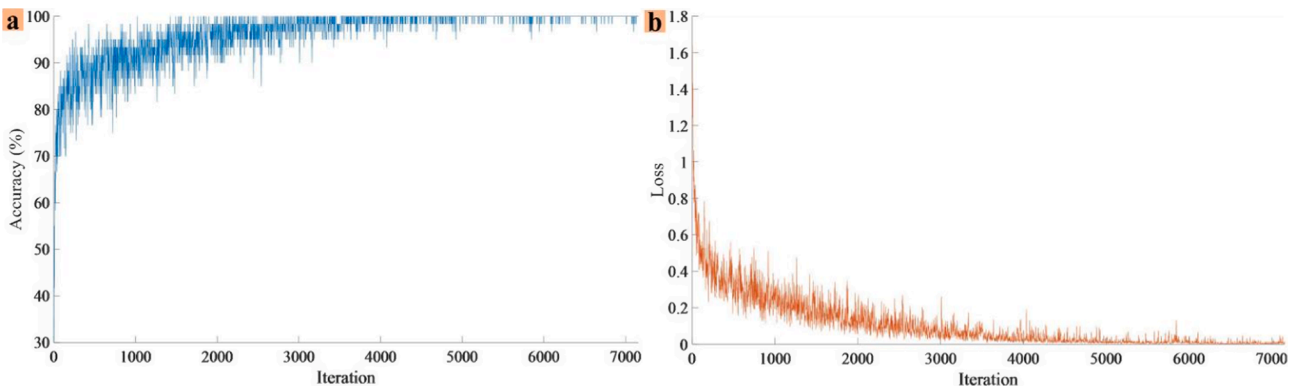


Fig. 9. Training process of Bi-LSTM model: (a) and (b) are respectively the accuracy curve and Loss curve in the training process.

of videos for that behavior in the test set, and *specificity* is the ratio of the number of videos with wrong recognition to the number of videos with other behaviors in the test set. *TP* and *TN* respectively denote the number of same videos with the current behavior pertaining to the other behavior recognition results and actual results. *FP* and *FN* respectively denote the number of different videos with the current behavior pertaining to the other behavior recognition results and actual results.

3. Results and analysis

The results of the behaviors recognition corresponding to the 1370 test videos, obtained using the proposed algorithm are shown in Figs. 10–12, where Fig. 10 shows the number of videos of the five behaviors in the test videos and the corresponding correctly recognized videos. Figs. 11 and 12 show the *accuracy*, *precision, recall* and *specificity* for the recognition of different behaviors.

Referring to Figs. 10 and 11, it could be seen that the highest recognition *accuracy* for the five behaviors was 0.985 for standing, mainly due to the obvious difference between the static standing behaviors and other behaviors, while the lowest *accuracy* was 0.950 for drinking. It was mainly due to the fact that the dairy cow's mouth may leave the water trough and move repeatedly, which makes it highly similar to ruminant behavior and easy to be misclassified.

In the 1370 test videos, the average recognition *accuracy* of the proposed algorithm for the five behaviors was 0.976. As shown in Fig. 12, the *precision* of the five behaviors ranged from 0.958 to 0.995, the *recall* ranged from 0.950 to 0.985, and the *specificity* ranged from 0.974 to 0.991, while the average *precision*, *recall* and *specificity* were 0.971, 0.965 and 0.983, respectively. After testing, the proposed algorithm took 2.751–3.684 s for feature extraction and 0.609–0.894 s for behaviors recognition. The average time for feature extraction and behavior recognition was 2.958 s and 0.712 s, respectively. The results indicated that the proposed algorithm could suitably recognize basic behaviors of the dairy cows.

4. Comparison of different feature extraction networks

In recent years, the VGG, ResNet, MobileNet, DenseNet and other advanced networks have rapidly emerged, and these networks are widely used in image feature extraction (Bychkov et al., 2018; Ao et al., 2019; Sandler et al., 2018). In this study, VGG19, ResNet18, ResNet101, MobileNet V2 and DenseNet201 were used to extract the feature vector sequences of the test videos. The behaviors recognition results based on the different feature extraction networks were compared to verify the effectiveness of the VGG16 used in the proposed algorithm.

First, the VGG16 in CNN-LSTM was replaced by the above six networks to extract the feature vector sequences of the test videos, and then the classification model of the head of CNN-LSTM algorithm was trained based on these features. Finally, the classification models of different feature extraction networks training were tested by the test set. The test results are shown in Fig. 13.

The accurate recognition of dairy cow's behaviors could not only provide a reference for the physiological health assessment and remote diagnosis, but also contribute to realize precision farming. The accuracy of behaviors recognition based on our proposed algorithm, VGG19, ResNet18, ResNet101, MobileNet V2, and DenseNet201 was analyzed in this study. The test results showed that the average recognition *accuracy* based on these 6 networks was 97.59%, 97.51%, 95.62%, 95.40%, 94.74% and 95.25%, respectively, while the variance of the recognition *accuracy* of these different feature extraction networks for the five basic

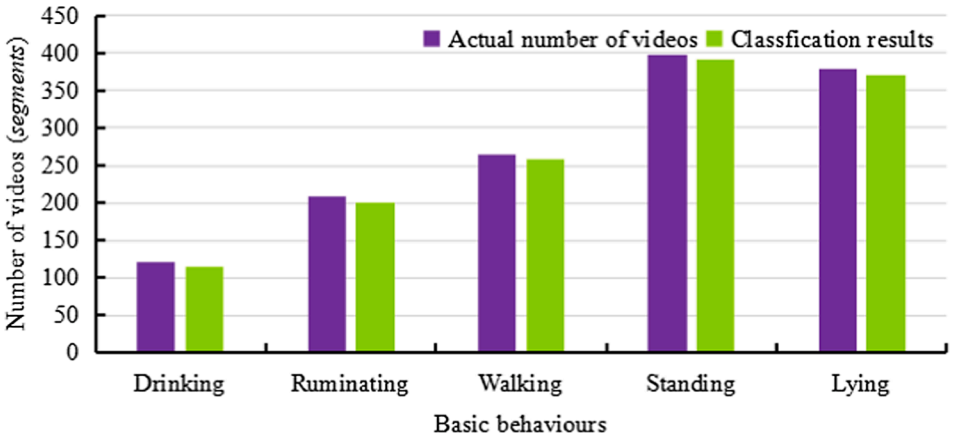


Fig. 10. Behaviors recognition results of the test videos: blue histogram and red histogram represent the number of real videos and the number of correctly recognized videos, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

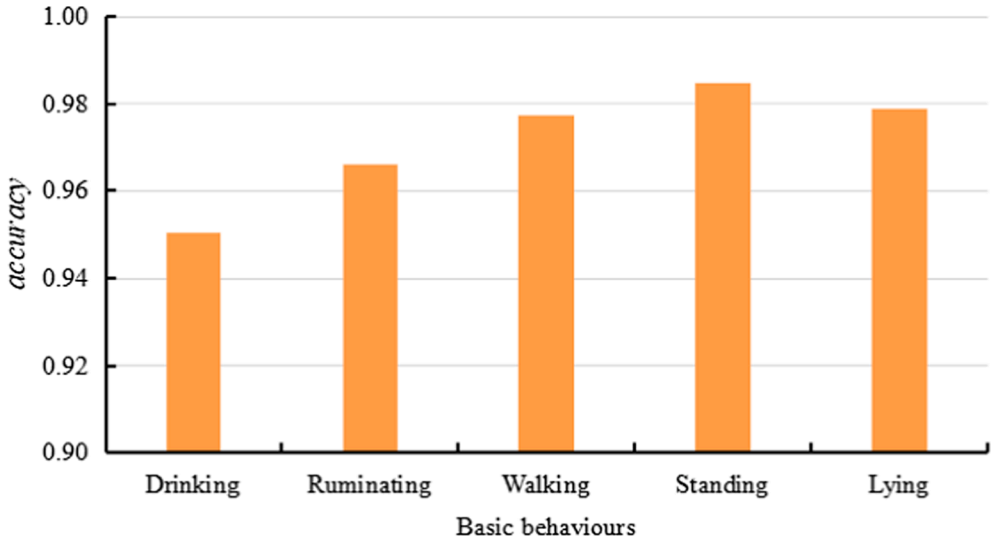


Fig. 11. Recognition accuracy of the proposed algorithm for different behaviors.

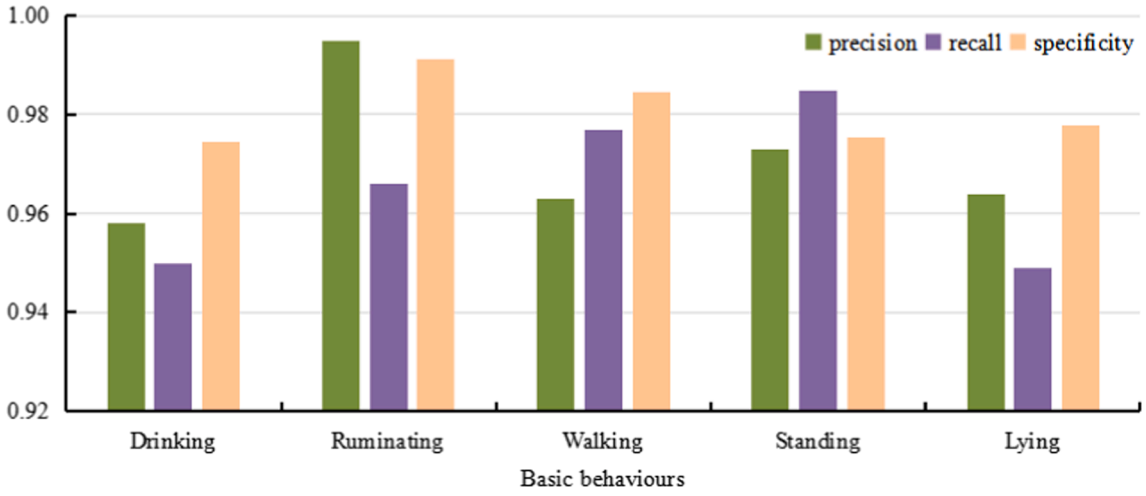


Fig. 12. The *precision, recall* and specificity of different behaviors recognized by the proposed algorithm.

behaviors (drinking, ruminating, walking, standing, lying) was 1.84×10^{-4}, 1.83×10^{-4}, 5.04×10^{-4}, 7.35×10^{-4}, 2.42×10^{-4} and 6.37×10^{-4}, respectively.

Fig. 14 shows the comparison of behavior recognition results between our method and VGG19. According to the analysis of test results, in the aspect of behavior recognition effect, although the method based on VGG19 relied on its deeper network structure and feature extraction ability, which made the variance of behavior recognition slightly smaller than that of the proposed method, the average recognition *accuracy* of the proposed method was 0.08% higher than VGG19 and the network parameters were about 5.31 MB lower than VGG19, indicated that our method has higher cost performance.

The recognition *accuracy* of lying behavior based on the ResNet18, ResNet101 and DenseNet201 was 0.57×10^{-2}, 0.67×10^{-2} and 0.99×10^{-2} higher than that of the proposed method, respectively. In addition, the rumination behavior recognition *accuracy* based on the ResNet101 and DenseNet201 was 1.43×10^{-2} and 0.11×10^{-2} higher than that of the proposed method, respectively. This was mainly due to the introduction of shortcut connections in the ResNet and DenseNet network structures, which effectively improved the features extraction and dissemination capabilities in these deep networks (Huang et al., 2020). Furthermore, the average recognition *accuracy* of the drinking behavior based on the ResNet101 and MobileNet V2 was 0.84×10^{-2} and 0.72×10^{-2} higher than that of the proposed method, respectively. However, overall, the proposed algorithm exhibited the highest average behavior recognition *accuracy*, with values 0.08×10^{-2}, 1.97×10^{-2}, 2.19×10^{-2}, 2.85×10^{-2}, 2.34×10^{-2} higher than those of the five comparison methods (VGG19, ResNet18, ResNet101, MobileNet V2, and DenseNet201), respectively. In addition, the proposed algorithm exhibited the smallest variance. These findings showed that our method was effective for the feature vector sequence extraction of the dairy cow videos in this study.

5. Discussion

In the complex and highly variable dairy farming environment, many interference factors exist, and the image quality of the monitoring camera is inferior. To verify the robustness of the proposed algorithm, several main interference factors that may affect the behaviors recognition were analyzed to illustrate the effectiveness of the proposed algorithm.

(1) Comparison of the proposed algorithm with the state of art researches

As shown in Table 2, it is a comparison of the main contents of

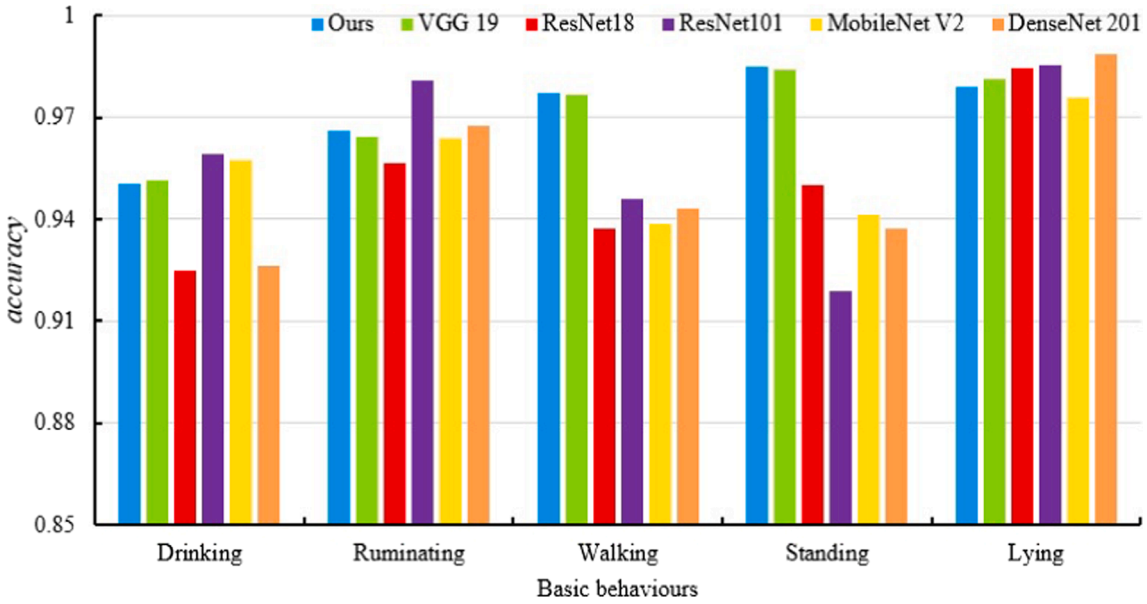


Fig. 13. Behaviors recognition result based on different feature extraction networks.

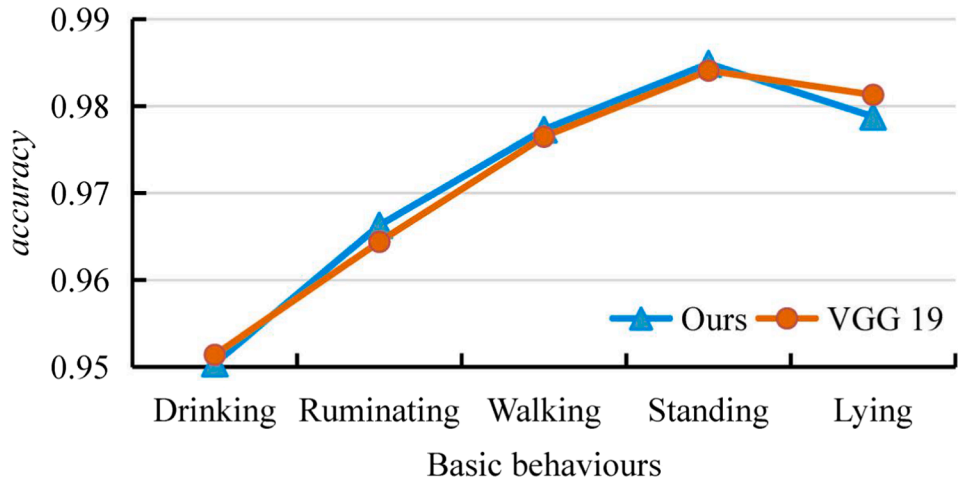


Fig. 14. Behaviors recognition result of our method and VGG 19.

related research on behavior recognition of dairy cows. Kiani (2018) applied decision tree classifier to identify standing, walking and lying behaviors of cows on the basis of collecting leg acceleration data. Compared with the data acquisition method of Smith et al. (2016), which fixed a collar on cow's neck, the movement acceleration could be measured more directly and effectively. To avoid the limitation of a single sensor and improve the robustness of the behaviors recognition method, Wang et al. (2019) collected the acceleration and position information of the cow's leg and realized the recognition of 7 kinds of behaviors by using the proposed integrated classification algorithm, with accuracy ranging from 0.80 to 0.99. Furthermore, Chelotti et al. (2020)proposed a method based on the recognition of jaw movements from sound, which were then analyzed by groups to recognize rumination and grazing bouts, and the recognition result was higher than 0.75. Although the touch-based sensors had better accuracy and robustness, they were easy to cause cow stress behavior, which made the sensors vulnerable to damage. Therefore, there was still some resistance in the practical application of recognizing dairy cow's behaviors based on contact-based sensors.

In regard to video recognition, Tsai et al. (2020) solved the problem of cow drinking behavior recognition by using an embedded imaging system, with an accuracy more than 0.90. To detect the estrus of dairy cows, Guo et al. (2019) used support vector machine (SVM) to recognize the mounting behavior after extracting geometric and optical flow features of the target area. The results of 30 tests showed that the accuracy of this method was 0.98. However, the above methods could only recognize a specific behavior, and most of them need to design feature engineering manually and the process was relatively complex. He et al. (2016b) solved the problem of various behaviors recognition. Based on calf target detection and feature sequence extraction, a clustering algorithm was conducted to recognize behaviors (lying, standing, walking and running), and the accuracy was higher than 0.95. Furthermore, Guo et al. (2020) proposed a behavior recognition method based on deep learning, which did not need to manually design features, and realized the recognition of calf's scene interaction behavior by fusing background subtraction and inter-frame difference algorithm. The recognition accuracy of this method was between 79.69% and 96.85%. The common shortcomings of the above-mentioned computer vision-based methods were that all the experimental videos in this study were high-definition videos shot manually, and the behaviors recognition problem in complex environments such as poor image quality of farm cameras, night and rainy days was not considered, which may limit the

practicability of this method.

In order to overcome the above shortcomings and realize the recognition of various basic behaviors of dairy cows in complex environment. On the basis of constructing the video dataset for cow's behaviors recognition, the CNN-LSTM algorithm was proposed to recognize the basic behaviors of cows, in which the feature sequence of each test video was extracted by the VGG16, and then the Bi-LSTM classification network was combined to conduct behaviors (drinking, ruminating, walking, standing and lying) recognition. The test results showed that the average *accuracy* of the algorithm was 0.98.

This study could provide valuable reference for cow behavior recognition in complex environments such as low-quality surveillance video, complex illumination and weather changes, and also contributed to the physiological health monitoring of dairy cows.

(2) Analysis of the influence of the illumination changes on the behaviors recognition results

The test videos in this study were shot from 6:00 a.m. to 12:00 p.m., and the time span ranged from early morning to late night, which ensure that the illumination in the dataset changed considerably. In order to more intuitively analyze the influence of illumination on the behavior recognition results of the proposed method, some examples in the form of images would be displayed. Fig. 15a shows an example of a cow's standing behavior that cannot be accurately recognized at night, and Fig. 15b and c shows examples of correctly recognizing the standing behaviors.

As shown in Fig. 15a, the cow's standing behavior was incorrectly recognized as ruminating because the cow's body has few white speckles, and under dim illumination conditions, the cow appeared to be 'submerged' in the background, which made it difficult for CNN to extract effective time series features from the video, which greatly affected the behavior recognition result. Furthermore, the cow's eyes were brighter and closed repeatedly, which was similar to the movement rule of ruminating, making the behavior being wrongly detected as ruminating behavior. Compared with in Fig. 15a, in Fig. 15b and c, the cow target was not 'submerged' by the background in a large area, and the contour and speckles were relatively clear, which was more conducive to the feature extraction and behavior recognition.

To further explore the influence of illumination changes on the behavior recognition results of the proposed algorithm, 1370 test videos were used for testing (including 794 videos collected during the day and 576 videos collected at night). The test result showed that the average recognition *accuracy* in day and at night in the test set was 0.981 and 0.971, respectively, and the variance was 0.58×10^{-4}, which showed that the illumination change had little influence on the behavior recognition accuracy of the proposed algorithm.

(3) Influence of weather changes on behaviors recognition results

Some examples of basic behaviors on rainy days are shown in Fig. 16. Since the soil in the outdoor exercise area was loose, reflective puddles would be formed when it rained, which affected the the gray values of some pixels in the sequence frames and may affect the behavior recognition results. Furthermore, the outdoor exercise area was open and the drinking pool was in the center of the outdoor exercise area, when it rained, the cows remained in the cowshed except when they moved to drink water, which made videos of only cows drinking, ruminating, standing and walking in the outdoor exercise area could be collected in rainy days. As shown in Fig. 16 a–d, the behaviors of the dairy cows are correctly recognized.

The algorithm was employed to test the behavior recognition of dairy cows on the test set (including 994 videos on sunny days and 376 videos on rainy days). The results showed that the average *accuracy* of behaviors recognition on sunny and rainy days was 0.978 and 0.974, respectively, with a variance of 1.0×10^{-5}, which indicated that the algorithm could well recognize cow behaviors regardless of whether occurrence of rain. The main reason may be that although these reflective puddles affected the gray value of some pixels in the video, these pixels were almost static, which were quite different from those of cows, and would not cause great interference to the feature vector sequence extraction of dairy cow's videos and affected the behavior recognition results.

(4) Effect of wind on behaviors recognition results

In this study, 1370 test videos were utilized to test of the proposed algorithm. The test videos included environments involving several wind velocity, namely, windless (0–1.5 m/s), breezy (0.6–3.3 m/s) and gale (3.4–5.4 m/s) conditions.

In the 1370 experiments performed using the proposed algorithm, the average *accuracy* of behaviors recognition under windless, breezy and gale environments was 0.987, 0.974 and 0.967, respectively, with a variance of 1.1×10^{-4}. The results indicated that the wind velocity did not considerably influence the behaviors recognition accuracy, and the proposed algorithm could recognize behaviors of the cows with a high accuracy regardless of the wind conditions.

6. Conclusion

To realize the intelligent recognition of the basic behaviors of single target cow in complex environments such as those involving inferior video quality and illumination variations, CNN-LSTM an algorithm for recognizing the basic behaviors of a single cow, was proposed. A total of 1370 videos were tested, and the main conclusions were as follows:

1) On the basis of extracting video feature vector sequence based on CNN, the proposed algorithm realized the recognition of a single cow's behaviors in a complex environment by constructing a Bi-LSTM classification model. When a cow was not 'submerged' by the background, the complex interferences such as low illumination, rain and wind velocity changes did not exert a considerable influence on the final behaviors recognition results, which indicated that the



Fig. 15. Examples of cow standing behavior recognition under different illumination: (a) standing behavior was incorrectly recognized as ruminant behavior at night, (b) and (c) standing behavior was correctly recognized at night.

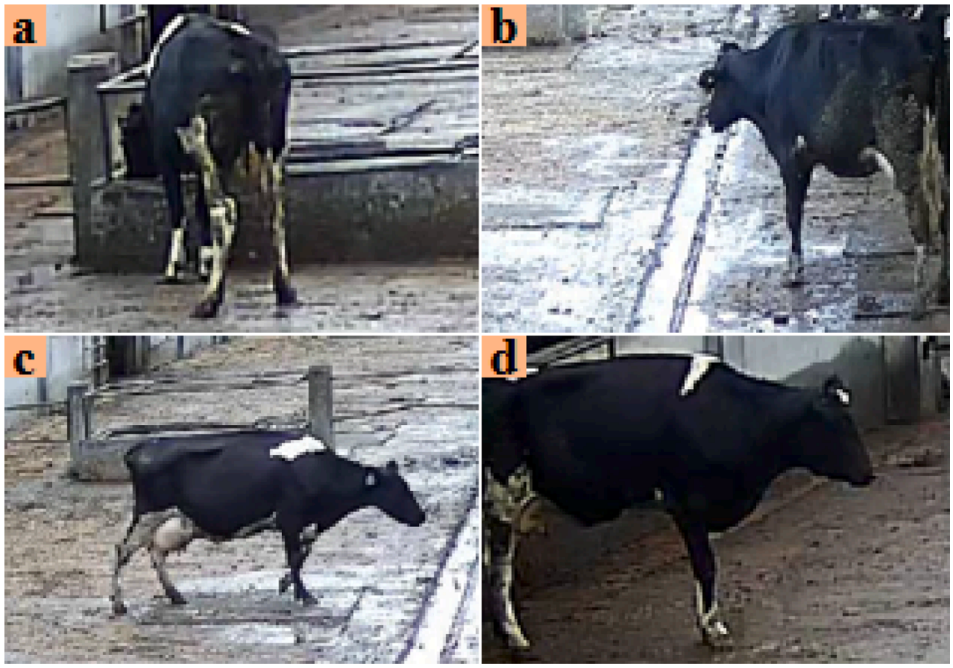


Fig. 16. Video examples of some basic behaviors on rainy days: (a) correctly recognize drinking behavior, (b) correctly recognize ruminating behavior, (c) correctly recognize walking behavior, and (d) correctly recognize standing behavior.

proposed algorithm could be used to recognize behaviors in complex environments.

2) The proposed algorithm could recognize a single cow's behaviors (drinking, ruminating, walking, standing and lying). Experiments pertaining to the behavior recognition in complex environment indicated that the average *accuracy* of the proposed algorithm was 0.976. This research could provide reference for the advanced behavior identification and understanding of multitarget dairy cows, and it was of significance for the physiological health assessment and remote diagnosis in the context of precision farming. The proposed algorithm was mainly based on deep learning and required high computing hardware. Moreover, this research focused on behavior recognition of a single target cow. In the future, we would be committed to combining target detection and model pruning technologies to compress the model as much as possible under the premise of ensuring performance, and to more automatically realize the behavior recognition of multiple target dairy cows.

CRediT authorship contribution statement

Dihua Wu: Conceptualization, Methodology, Software, Writing - original draft, Visualization, Formal analysis, Validation. **Yunfei Wang:** Resources, Writing - original draft. **Mengxuan Han:** Resources, Writing - original draft. **Lei Song:** Resources, Writing - original draft. **Yuying Shang:** Resources. **Xinyi Zhang:** Resources. **Huaibo Song:** Writing - review & editing, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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